COMPLETE RESEARCH RETROSPECTIVE

Latent-Space-Reasoning

Every research arc, theorem, validation, refutation, and open obligation of the program - from the founding denizen question through five complete theorem families and the dimension-asymptotic universality law.

> Six campaigns. Eleven adversarial reviews. 1085 ledger entries.

REPORT

LATENT-SPACE-REASONING: THE COMPLETE RECORD (December 2025 - September 2026)

The sole surviving artifact. 1,540 git commits across ten months, ~1,085 ledger rows in the modern era, 50+ adversarial audits in the classical era plus 14 in the modern. Every campaign, theorem, number, failure, and lesson - written for a reader with no repository access.

PART 0 - THE TRUE ORIGIN: THE EVOLUTION ENGINE ERA (December 2025 - May 2026)

0.1 The founding idea

The project began (December 8, 2025, initial release) as a Latent Space Reasoning Engine: treat a frozen model's latent generation path as an editable, inspectable search space. Target loop: sample -> inspect -> repair -> aggregate -> verify -> realize. The first era (~200 commits through May 2026) built an evolutionary search over latent perturbations:

- Evolutionary latent search with survivor decay, score caching, diversity features to prevent collapse (V2-V10 verifiable evolution iterations with review-driven redesigns).
- Hyperbolic latent geometry: GeometryConfig, HyperbolicMutation on the Poincare ball, HyperbolicCrossover via Karcher-mean barycenter, MixtureCurvatureMutation with per-individual curvature co-evolution, Poincare CMA-ES (low-rank covariance adaptation on the Poincare ball).
- Information-geometry steering: dual steering module with Newton logit steering.
- Soft-prompt experiments with attention-masking interventions for the attention-sink hypothesis test.

0.2 The perturbation paper (the era's flagship)

Prepending random embedding-scale tokens to a 4-bit quantized Qwen3-4B produces a sharp, non-monotonic improvement in chain-of-thought arithmetic: two random tokens raise mean accuracy from 32% to 51.6% (+19.6 pp) across ten independent directions; one token 42.7%; three or eight revert to ~44%. A first-token logit probe confirms perturbation does NOT activate reasoning mode (already saturated >99.99% under thinking-enabled decoding) - it modulates the downstream trajectory. Different directions solve different task subsets: ten two-token directions collectively cover 100% of tasks vs 80% (three-token) and 92% (eight-token); the three-token oracle is a strict subset of the two-token oracle (over-perturbation contracts the reachable task space). Cross-model: Qwen3-8B 8-bit +12.8pp (oracle 80%, McNemar 16/0), DeepSeek-R1-Distill-1.5B (oracle 100%, McNemar 6/0), phi-2 +6.7pp out-of-family. A paper draft carried the result with oracle coverage-vs-budget as the main figure.

0.3 The era's hard lessons

- STRONGLY EXPLOITABLE finding logged honestly during soft-prompt work: 32% accuracy range across random latents (p=0.006) - the latent scorer itself was gameable, seeding the later rule that automated scorer numbers are evolutionary guidance only, never quality evidence.
- Verbosity-inflation artifact caught and fixed (response/response_raw bias) - the template for later artifact discipline.
- The era closed when the nested-arithmetic capability claims were WITHDRAWN (Part 0.5).

PART 0.5 - THE DIFFUSION REASONING PROGRAM (May - August 2026, ~330 commits)

Between the engine era and the native-math pivot sat a substantial diffusion-language-model program:

- DIFFUSION LATENT REPAIR (the promoted public result): fixed denoise 0.412 -> latent repair 0.531 at 2.6x GPU cost (lower-cost point 0.509 at 2.4x) on the lean mixed benchmark.
- MULTI-LATENT AGGREGATION v10/v11: v10 (48 fresh tasks) 40/48 coverage, 40/8/0 W/T/L, lift +0.077, zero contradictions; v11 (96 tasks, LLaDA-only) 87/96, lift +0.100, Wilson lower 0.831 - all 13 frozen gates passed with 2x replication. But the rubric was gameable (keyword audit RED).
- BLINDED PAIRWISE (N=50): task-specific clauses preferred 33/50 (66%, p=0.016) - confirmatory GO.
- V12 FILTERED REPLICATION (116 fresh tasks): NO-GO on the Gemini judge family - generic clauses BEAT task-specific (63 vs 53); same-vendor confound (generator and judge both Gemini).

 SEPARATRIX PROBE: interior correctness islands in 26/35 tasks along wrong-to-correct interpolation paths; bifurcation analysis found only ODD transition counts (a geometric constraint: wrong-to-correct requires odd crossings), mean 3.0 transitions/task; exploratory only (determinism, endpoint-selection, and norm-preserving controls lacking).

 The diffusion theory effort (30K chars of half-formal theory; claim ledger; ~23 numbered value-model/probe versions including span probes, retention gates, KV tomography, counterfactual probes, phase measurements) - archived when the program pivoted; retained as historical evidence.

 HARDWARE-DEPENDENT DECODING DETERMINISM (surviving finding): GH200/CUDA-13 runs differ on 2/5 sequential repeats; 8 byte-identical batch rows give 13/40 distinct completions; RTX 5090/CUDA-12.8 bit-identical across three processes. Embedding perturbation is a causal diversity source ONLY on deterministic stacks - any diversity claim must report the stack's noise floor.

 THE NESTED-ARITHMETIC WITHDRAWAL (August 2026, the era's terminal correction): the "perturbation unlocks capabilities that scaling cannot" claims were withdrawn after independent reanalysis showed 76-100% of every published rung was THINKING-MODE TRUNCATION, not arithmetic. --no-think at the same cap takes Qwen3-32B from 0% to 100% and every rung 1.7B-32B to 96-100%. What survived: under a binding cap, perturbation shifts trajectories toward completion (100/100 completed-correct; termination 24%->38%) - irrelevant on arithmetic where no-think is cheaper and better. The correction record became the template for every later correction: full public record, claim map, ground-truth index.

PART 0.7 - THE NLM ROUNDS: THE FIRST NATIVE-MATH ATTEMPT (August 27-29, 2026, 37 rounds, 22 audits)

After the native-mathematics direction was set, the first measurement program ran as numbered rounds with formal audits, on DINOv2/Qwen3 chart worlds:

 NLM-001 (lexical next-token-KL instrument): BOUNDED NEGATIVE FALSIFIER - native calibration KL did not beat a learned diagonal metric on contextual hidden states for held-out closeness orderings (Qwen3-0.6B Delta = -0.058 [-0.22, +0.03]); a taxonomy leak in the authored probe set was caught and removed.

 NLM-003 (R-metric vs F-Fisher): the apparent R win (0.734 vs 0.630) was a TAXONOMY LEAK - the required sensitivity rerun removed PB_coarse and reversed the result (R 0.586 vs F 0.667). Claim withdrawn. Cosine and Euclidean still dominate both leak-free native candidates - the first clear statement that imported metrics were beating the native attempts.

 NLM-004 (random-init null): trained chart structure is real (cosine fine-label 0.946 trained vs 0.575 random-init; kNN 0.761 vs 0.069) and separate from cheap image statistics; but affine interpolation smoothness is an imported chart path, not native evidence.

 NLM-005/006b (transport/substitution in DINOv2): the label-preservation rule failed (crop50 0.458, invert 0.317, mix50 0.185, occlude50 0.416 - all OOD); non-diagnostic under its lock.

 NLM-007 (the terminal ladder, Rounds 13-34): a law-complexity ladder (mean, kNN k=1/5/20, ridge, low-rank affine, kernel ridge) on carrier-block hold-out folds with A/B punctuation sentinels. The full arc: endpoint adjudication, corrected slot rerun, cheap moot-makers, displacement tests, forward-time locality, LOCO controls, unseen words, residualization 2x2 - each round pre-registered, audited, corrected. TERMINAL VERDICT: closed under the allocation rule, not a scientific null - bounded F4-F20 condition robustness established within the pinned decoder/authored population; the raw token-context comparator was P_static-aligned; Round 34b INCONCLUSIVE in both sentinels (P+C minus P fails the redundancy STOP; C_perp retention fails). The identity-baseline withdrawal and the four-cell common-scale adjudication were the era's methodological anchors.

PART 0.8 - THE TOY QUOTIENT PROGRAM, HANDLE, AND THE FROZEN-RESIDUAL LANE (August 29 - September 2, 2026)

 ROUNDS 36-37 (toy operational-quotient worlds): 16 four-bit states with toggle/swap/no-op actions, identity defined solely by bisimulation. Round 36: both cells behaviour-ineligible for the exact reducer. 36b (behaviour-fit ladder): all cells ineligible. 36c (quotient-trained positive control): both cells FAIL. 36d (frozen-chart positive control): joint FAIL. 37 (presentation-duplicated 32-to-16): NO ARCHITECTURAL WIN. The toy program ended under the governance amendment: exact certificates are diagnostics only; real latent spaces only.

 FBA-0 (the algebra testbed, Z/8 x Z/4 POMDP, 6 architectures x 3 seeds): FAIL 0/3 on joint tasks - factorization accuracy without joint composition. K7b interchange passed for single slots.

 HANDLE-mu (handle-addressable latent dynamics, the co-developed world): Rung 1 FAIL (status F1 0.33 vs 0.90 - MSE loss drowned the 4-dim status signal; flat GRU worse than historyless; events predictable from the current frame alone). V2 (rebuilt with separate CE head, latent keys): dense-historyless gap tripled to 0.075, latent keys working as designed - but two independent reviews converged on the freeze verdict: HANDLE-mu is calibration, not the artifact; the tunnel pattern named for the first time ("propose algebra, build world, fail, repair evaluator, interpret calibration as progress").

 RCQ-0 (real causal quotient on Finch-3B/RWKV): entity discrimination TV=0.43 PASS, state replay bit-exact, entity swap confirmed (TV=0.41/0.46) - but composition fails; RWKV below 3B lacks entity discrimination. Closed administrative/protocol.

 PFC-0 (path-fiber calculus): TASK_POPULATION_VOID - K is a researcher-imposed 3x3 stochastic matrix; its composition is ordinary matrix multiplication (distance-from-claim 2: imported probability algebra, not native).

 The frozen-residual lane closure (OCI/RAC): the sentence-position routing (95.8% B20 transplant), structural routing (96.5%), and crossed-factor steering results - 99.1% explained by a logit-additive null through the linear unembedding; NO_INTERFACE for template-invariant response fibers. Everything found reduces to factored encoding (PCA) plus logit addition. CLOSED.

 coordinate_v1/v2/v3, interchange_v1/v2, state_bus_v1r1, control_cost_v1: all closed with registered fails or invalid baselines. The never-say lists (audits 27-31) are binding on any citation: never say "a two-dimensional latent grammatical coordinate was discovered", "the state persisted through generation", or "frozen pretrained residual streams do not contain native mathematics" (the last is the underclaim guard).

PART A - THE MISSION AND ITS EVOLUTION

A.1 The founding question (August 2026)

Mathematics was invented by inhabitants of a world to navigate it - counting, measuring, mapping, predicting - and its laws were shaped by what that world made necessary. Invert the dynamic: take the latent space as the world. Ask what a denizen of that world would have to invent to find its way - what counts as the same place, what a move is, what effort a move costs, what a map is, what regularities make prediction possible - and let that need decide which primitives and laws get built.

The standing adversarial check that shaped everything: every empirical tool a researcher naturally reaches for (PCA, Procrustes, cosine similarity, Euclidean distance, orthogonal rotations) is R^n mathematics applied to a space whose premise is that it is not R^n. A result demonstrating linear algebra working on embeddings is activation engineering, not the goal. The bar: what mathematical structure does the latent space have that R^n does not describe?

A second lens, held throughout: structural properties that make a space hostile to structured reasoning are first-class results ("holes"). A proven hole identifies what a next-generation latent space would have to change.

A.2 The direction pivots (in order)

1. Native mathematics of LLM latent spaces (August-September 2026): denizen primitives discovered behaviorally in frozen small models.

2. Cumulative capability acquisition (S1-S65, September 13-14): can a system combine useful computations from successive pretrained donors while retaining prior capabilities? Paused at user request; no acquisition established.

3. General mathematics of high-dimensional spaces (September 17 onward, the current mission): LLMs are too expensive on the available hardware (sustained GPU load hard-crashes the laptop; CPU-only) and carry too many confounds per unit of insight. Latent spaces demoted to one later instance of the foundation. The axiom, never in question: mathematics behaves differently in high dimensions; concentration of measure, almost-orthogonal packing, interference, and distance uniformity are the native physics.

A.3 Terminal conditions (standing orders)

Work continues non-stop until either (a) a proof-level demonstration that the program is a dead end, or (b) multiple independent adversarial reviewers confirm a contribution at the level of a Nobel Prize or Fields Medal. Neither has been met. The program is paused mid-arc at the owner's request with all records current.

PART B - CAMPAIGN 0: THE LEGACY PROGRAM (pre-August 2026)

The prior program studied latent perturbation and diffusion repair. It closed when its nested-arithmetic benchmark claims were shown to measure termination-under-a-token-cap rather than capability. Its transferable residue: the benchmark-correction methodology itself (measure termination separately from accuracy; a capability claim void without it), retained as a hard evidence gate for every later campaign: termination rate reported beside every accuracy; the cheapest direct remedy run as an arm before any scaling comparison.

PART C - CAMPAIGN I: NATIVE LATENT MATHEMATICS IN REAL MODELS (August 27 - September 6)

C.1 Phase 1 - the R^n trap discovered (50 experiments/audits)

Operational quotients, coordinate/interchange/state-bus/control-cost constructions, one-write memory staircases, constructed register substrates - all closed. The three converging findings written into the record: (1) the R^n trap - the program could not escape its own measurement apparatus; (2) no new mathematics found - every proposed native object mapped to established mathematics (coalgebra, predictive state representations, automata, causal abstraction); (3) architectural analysis - transformer KV-cache state is non-R^n but inaccessible from the fixed-d residual stream (the Robust Port-Compression Conjecture); SSM fixed-dimensional state is accessible but described by standard spectral theory. Terminal experiment PSQ-3a: NO_INTERFACE (69.14% accuracy vs 95% gate; task-trained LoRA could not establish the interface).

The deposits that survived: theory/AXIOMS.md (D1-D9: future-response geometry - identity as zero future-response distance, nonexpansive legal moves) and the Nine Breakpoints (see C.6).

C.2 Phase 2 - the observational ceiling (18 experiments, Qwen3-0.6B micro-world)

Commitment bottleneck, non-naturality of probes, fusion-fission distributed mechanism, QPC-1 hole result. Structured negatives only; the micro-world could not support native-structure claims.

C.3 The SVB program and the settling-time law

Method: Scope-Variable Binding - Python lexical scoping as a probe; nested def blocks create scope depth 1-4; the model must report the value at the outermost scope; observable sigma = P(correct outer value) over 11-bin readout.

- SVB-0 (Falcon-H1-1.5B, 1854 calls): INSUFFICIENT_SCOPE_BINDING (two-var 76.7% below 80% gate; single-var sigma=0.68, kappa=0.71, iota=0.36) - signal 100x stronger than the entity-location family.
- SVB-1: SETTLING TIME LAW - sigma decays with depth (0.68, 0.50, 0.28, 0.23) but neutral suffix tokens INCREASE sigma at d>=2: gains +30% (d2), +57% (d3), +88% (d4). Optimal settling sigma* = (0.68, 0.65, 0.44, 0.43): raw depth curve overestimates binding loss ~2x at d4. Geometric-decay prediction rejected (non-constant ratios).
- THE FALCON-H1 MAMBA CACHE BUG (critical instrument correction): HuggingFace's multi-token cached continuation hardcodes zero initial SSM state (line 819 modeling_falcon_h1.py) and omits cached conv state - ALL suffix-algebra experiments on Falcon-H1 used the defective path. Verified by one-shot vs two-shot TV=0.118 vs single-token-decode TV=0.000000. Corrected measurement: tokenwise cached query processing within 1.2e-6 TV of full text; legacy multi-token path 0.546-0.885 TV. Qwen3 unaffected (different code path).
- The surviving Qwen3 finding: the depth-dependent suffix effect (see C.5).

C.4 The suffix-action algebra (Codex-designed decisive experiments)

Words over generators {C (comment), P (pass)} act on readout distributions. Central question: does the action monoid collapse to a finite quotient (band, right-zero, etc.)?

- LRB decisive experiment: H-LRB REFUTED. R(C,P)=0.091 and R(P,C)=0.141, both lower bounds above the 0.06 epsilon (stratified bootstrap 10K, seed 42). Non-collapse confirmed: N(C,P)=0.091.
- Band2 experiment (12 arms incl. CPCP, PCPC, CPP, PCC): the monoid does NOT collapse to a finite quotient; branch coalescence not absorption; CPCPC differs from CPC (TV=0.099); sigma degrades CPC=0.486, CPCC=0.439, CPCPC=0.393.
- EMI (execution-mode invariance, Codex-designed 4-mode L/W/G/F spec): replay calibration delta=0.0 (perfectly deterministic instrument); mode L bit-for-bit reproduces historical data; query-continuation defect verified: tokenwise cached query processing within 1.21e-6 TV of full text. All suffix-algebra conclusions scoped to the verified cached pathway.
- Zero-call analyses: common-tail forgetting REFUTED (shared tail length has no predictive power, Spearman=0.059); word-space geometry - pairwise mean-TV is a valid metric (0 triangle violations/4095 checks), 2D MDS captures 93% variance; C/P generator asymmetry - C absorptive (C-tail cluster TV 0.05-0.09), P disruptive (P/PP isolated at TV 0.24-0.30) - not predicted by any symmetric generator model; string baselines fail (edit distance rho=0.077, length diff rho=0.311).
- Final status: native object redefined as a Moore-style response automaton; no stable operator algebra on the response simplex (lumpability high R2=0.79-0.98 but composition amplification 2.6-13x breaks it; a scalar logit-bias model fits single-suffix data as well).

C.5 The depth-dependent readout effect and its mechanism program

One fixed appended comment (# No changes.) raises mean correct-value probability by up to 12.6 pp at nesting depth 4 (Qwen3-1.7B-Base; sigma d1-d4: 0.958/0.897/0.792/0.753 without, 0.970/0.945/0.900/0.880 with). The full follow-up program established:

- Content contrast exceeds variable-mention contrast 6.3x (2x2 crossover) - variable mention alone ruled out on that panel; responses still differ between paraphrases.
- Single-count maximum: among counts {0,1,2,4}, count 1 maximizes sigma at all depths d2-d4.
- Surface-specific not role-specific: within-class CV 0.33-0.51 across same-class paraphrases.
- Order effects positional not compositional (A-then-M vs M-then-A median TV 0.068; scalar gain fit absorbs it).
- Neutral continuations keep presentation pairs close (all 32 within 0.01 TV) while a reassignment announcement separates 29 of them.
- Mechanism unresolved: lexical cueing, added computation, attention-state change not separated. The honest claim: suffix-conditioned readout is a real inference-control phenomenon in this model family; the panel is deferred.

C.6 The Nine Breakpoints (design constraints for native mathematics)

1. Presence =/= causation (linear probes find ghosts: strongly decodable concepts with zero causal effect). 2. Single-site =/= distributed (factual recall shows whole-state dependence). 3. Vector distance =/= semantic distance (cosine-close points behaviorally different). 4. Fixed dimensions =/= fixed structure (context-dependent effective dimensionality). 5. Vector composition =/= computational composition (composition happens in the forward pass, not vector arithmetic). 6. Observation =/= state (choosing what to probe constrains what can be found). 7. Snapshot =/= computation (layer-l representation needs the trajectory). 8. R^n tools find R^n structure (PCA finds linear structure because PCA IS linear structure). 9. Metric blindness to composition (four fact-worlds with cosine ~1.000 produce different behavioral outcomes under intervention).

C.7 The write-boundary line (CR-10 to CR-16, the continuation-refinement calculus)

Central question: if a model had an internal replace-this-fact operation, would compositional reasoning become possible because dead history is truly dead?

- CEG-1 (custom 4D attention masks, Qwen3-1.7B, 914 forwards): competence preflight FAILED (83.6% < 95%); the binding design required NO_INTERFACE. Licensed sentence: bounded attention-mediated context dependence, not semantic relay of dead information.
- CRC-0/CRC-1: both identity partitions (Q=32 distinct rows); CRC closed as a discovery engine on Qwen3-1.7B. Retained as downstream assay only.
- CR-15 (adopted): exact selective overwrite has a lens structure (preserved context x store); write orbits from one initial place bounded by the product of register cardinalities; eta-accurate transition-closed realizations bound equal-endpoint histories by 2*eta under every registered continuation; a finite behavioral budget N bounds lifetime error by (N-1)*epsilon - sharp. A trained 17K-parameter overwrite cell realized perfect fact replacement on its training panel without being a contraction - closing the coordinate-certificate route.
- CR-15.8: min{1,(N-1)epsilon} lifetime bound without registers or normal forms. CR-15.9: the register-free congruence bound. CR-15.10.6: the modulus omega_a^W(t) as the organizing quantity for interface-relative interchangeability (countdown omega=1 at every finite depth; doubling omega=min{1,2t}; selective-write theta_t on the addressed fiber; decaying nuisance (1/2)min{t,delta}).
- CR-16: fixed-token causal site invariance retained; exact two-route decomposition withdrawn. The analytical cycle stopped per audit #41 ("not working as a route to a real-model artifact" - the mathematics retained as transferable constraint).
- The final scoped negative (Block 3, multifiller panel): H and matched successor fillers violate the 0.002 length-only prediction on 24/24/23 roots - generic readout sensitivity unexcluded.

C.8 The axioms adopted (D1-D9)

Future-response geometry: identity as zero future-response distance; nonexpansive legal moves; surgeon-vs-denizen refinement; the Robust Port-Compression Conjecture (transformer KV-state inaccessible from fixed-d residual stream once prefix complexity exceeds d; unaudited open deposit, close to known memory lower bounds). Standing caveat from the audits: this is established behavioral mathematics, not yet discovered native vocabulary; value must come from effective predictions about a real model.

PART D - CAMPAIGN I-CONT: THE BEHAVIORAL TRANSFER PROGRAM (the 86-experiment foundation)

This program (spanning late August-September) characterized how code perturbations affect model output across 10 language models - the empirical foundation on which the HD-math campaign later built. Every claim here carries its adversarial-review corrections (R12-R32).

D.1 The perturbation-sensitivity hierarchy

86 experiments, 69 cells x 24 perturbation types, full-logit and 11-bin readouts, models: Qwen3 (0.6B/1.7B, base+instruct), SmolLM2 (135M/360M/1.7B, base+instruct), Phi-1.5, StableLM-2-1.6B, CodeGen-350M. All CPU, deterministic, fully logged.

- WHERE you modify code is the dominant factor: after-expression edits (comments, whitespace, dead code) produce consistent reliable effects (AA ~ 0.7-0.9); expression-adjacent modifications produce heterogeneous cell-specific effects (DD ~ 0.1-0.5). Universal across tested models (scoped: observed on tested panels, not established universality).
- The 2x2 D-reliability diagnostic: D unreliability is STRUCTURED (within-op rho 0.632 >> cross-op 0.366, 2 models) - 5-6 of 6 operations reliably measure different cell properties; own-line versions more reliable than adjacent (O-O 0.660 >> D-D 0.412 both models).
- R29 inline-inert experiment: WHICH-LINE EFFECT IS ZERO (I-I = 0.878/0.789, within noise of A-A). In-expression edits (J group) are INTERMEDIATE (J-J 0.53-0.59 vs I-I 0.87, D-D 0.29) - neither inert nor D-like; whitespace inside expression inert (0.79-0.86); single-operand parenthesisation partially non-inert (0.40-0.66).
- R31 spacing/paren (Qwen3-0.6B, 226s CPU, anchors reproduced to 4e-6): SURVIVING - p_double ((expr)) strat4=0.279 < d_parens 0.655 (paired diff -0.376, CI [-0.60,-0.15]). WITHDRAWN - per-operator mechanism, betwOp inversion, token-context refutation, rarity dead. LEADING ALTERNATIVE: header shielding (parens make the header redundant, depth-monotonically; on - mean dist 33.6 -> (e)=24.4 -> ((e))=14.0).
- R32: nesting floor at depth 2 ((((expr))) ~ ((expr))); all four operators show small mean distance INCREASE at depth 3 - floor real, not pure monotonic compression.
- The // disambiguation artifact: a // b with 2 spaces read as C-style comment by the bare surface (10x distance inflation); the header restores floor division. Surface disambiguates language. The %-disambiguation artifact is the same class.
- MEANING: on the tested arithmetic panels no separable meaning effect was detected (within-family: +0.002/+0.008/-0.090 across 3 architectures, all consistent with zero); meaning is confounded with answer change in this domain. Factor ordering (final, post-R30b): position >> operator/sign >> (form, answer, meaning non-identifiable below sign). Three-level plateau: outside-expression (0.77-0.85), AST-preserving in-expression (0.53-0.60), computation-changing (-0.01 to 0.57).

D.2 The multiplicative model and its deflation

d[cell, perturbation] ~ fragility[cell] x magnitude[pert], log-space R2 = 0.82-0.88. R21 deflation: median perturbation angle 0.04-0.28 rad (small-angle regime) - log-additivity is expected from smoothness alone (the first-order Taylor expansion); the CONTENT is the 12-25% residual, which is real deterministic structure. The residual is organized by surface features (edit locus, comment-vs-statement, token adjacency), not by meaning preservation (R21's central result: attenuation ratio 0.79-1.01 - a single common cell factor with column-specific noise reproduces the gap to within 0-20%; "not predictable from inert" withdrawn, "less reliable" supported).

D.3 The kill experiment (the gradient, position-controlled)

Qwen3-1.7B, fp32, 40 cells x 24 perts, pre-registered: same-type transfer a<->b = 0.811 (new inerts predict old inerts); position partial confound ~35% (a<->c = 0.611); the gap is REAL after position control (a<->d = 0.333, rewrites don't transfer); rewrites NOT a coherent class (d<->d = 0.224). Statistical significance p ~ 1e-13, 1e-8 (Mann-Whitney). Replicated SmolLM2 (compressed gradient 0.281) and Phi-1.5 (Q1 readout: gap 0.217, p=5.26e-11 after the trailing-space discovery). Phi readout saturation: removing ONE trailing space from the query flips Phi from code-continuation (P(digit)=0.001) to digit prediction (P(digit)=0.980, accuracy 95%) - tokenizer mechanism: Q0 ends on space token (newline prediction), Q1 ends on colon (digit prediction). Phi KNOWS the answer but outputs newline under Q0. Model-specific admissible readouts govern all 5-architecture comparisons.

D.4 Cross-model structure

- Magnitude (which change types matter): ~40% shared across models (perturbation-magnitude transfer rho ~ 0.397, driven by the shared C-group floor; within-family 0.678 vs cross-family 0.350). Qwen3 is the architectural outlier (d_temp_var ranked #1-2 vs #16 median elsewhere).
- Fragility (which inputs are vulnerable): ~0% shared. Within-family rho near zero (Qwen3-0.6B vs 1.7B: 0.032; SmolLM2 135M vs 360M: -0.123); two significant NEGATIVE cross-family pairs. Cells most fragile in one model are LEAST in another (1**5 rank 0 in Qwen3-1.7B, rank 35 in 0.6B). At n=40 underpowered (power ~50% at rho=0.3); needs n>=200 diverse cells to be testable.
- Instruct effect: gap direction model-specific (Qwen3 up 1.5x, SmolLM2 down 0.89x) - architecture-dependent, n=2 pairs. Qwen3-0.6B instruct destroys D-group coherence (0.394 -> 0.169) while SmolLM2 preserves its fragility fingerprint (rho=0.689).
- 11-bin ~ full-vocab for code domain (TV ratio 1.0004, identical flips) - no measurement artifact concern; the 11-bin readout is a code-specific optimization.

D.5 The Elhauge-swap lesson and task degeneracy (adversarial reviews 10-11)

A recurring cheapest-confound pattern, named by the reviewer: a task that is degenerate (unigram targets, droppable tails, construction-forced contrasts) explains "universal" results without any general high-dimensional principle. The program's response discipline: before claiming any class-level law, test whether the construction itself forces the conclusion - this discipline carried directly into the HD-math campaign's reviews.

PART E - THE OMEGA MODULUS PROGRAM (~28,000 readouts, 5 architectures)

Question: the modulus of continuity of behavioral distance under appended continuations - omega_a^W(tau) = sup{d_W(T_a x, T_a y) : d_W(x,y) <= tau} on finite panels of 11-bin readout distributions.

- CROSS-ARCHITECTURE CONFIRMATION: three families show amplification (Qwen3 0.6B/1.7B/4B, StableLM-2-1.6B, Gemma-3-1b); Pythia-1.4B shows zero (insufficient code training). Amplification pattern model-dependent (Qwen3 ranks identity highest; StableLM neutral; Gemma pass).
- SCALING: identity amplification monotone within Qwen3: 0.003 (0.6B) -> 0.014 (1.7B) -> 0.051 (4B).
- GENERALIZATION AXES: cross-variable 83-92%; cross-pair LOO +27.5pp; cross-query format 100%; cross-code-structure 60-80%; perturbation diversity 14/14 novel types amplify; rank-based cross-pair Spearman 0.52.
- HELD-OUT PREDICTION: omega trained on standard continuations predicts novel perturbation types at 83.3% (+33pp over baselines; Cohen's d=3.49, permutation p=0.0004).
- NL OMEGA (full-vocab TV, exps 32-45): the 11-bin projection concealed NL differences by factors 18,000-64,000x; with full-vocab TV, NL amplification (5-17x) reaches code-range levels (passive-voice pair: question mark amplifies 16.9x).
- LAYER PROFILES (1.7B): omega monotonically increasing through depth; first nonzero at L20 (0.010, 100x jump from L16-19 at 1e-5); final 0.049; 0.6B onset EARLIER (L8) but weaker final - larger models concentrate variable-tracking in late layers.
- QUANTIZATION (exp 34): INT8 reshapes the omega profile as dramatically as base/instruct differences; mean TV(fp32,int8)=0.378. Omega is precision-dependent.
- THE OBSERVED STEP (exp 31): sharp jump at the first identity statement 0.003 -> 0.047; non-monotone with repetitions (rep_2 < rep_1, interference).
- REPETITION DISENTANGLEMENT (exp 40): pair-dependent with OPPOSING directions - strong pairs additive (II>I, last-wins NI>IN, neutrals accumulate NNN>NN>N); tight pairs OPPOSITE (suppressive II<I, first-wins IN>NI). I_paraphrase dramatically different from direct I (0.001 vs 0.022) - syntactic form matters.
- AUTHORITY FRAMING (exp 43): "# According to the documentation:" before neutral SUPPRESSES omega 97% on tight pairs (0.034x) but AMPLIFIES 3.5x on wide pairs.
- CONSEQUENCE PREDICTION (exps 49/53): cross-surface TV predicts argmax flips at AUC 0.918 (eligible-only); single-surface top-2 margin gets 0.80-0.86 - 80% of signal from one surface. On BASE model: weak (F1=0.129). The 1.7B sits in the vulnerable-but-capable sweet spot; 0.6B too weak (2.3% accuracy); 4B too robust (91.2% accuracy, 2.8% flip rate).
- CROSS-MODEL: tau=0.3 does NOT transfer across sizes (both 0.6B and 4B have F1=0.000 at that threshold).
- THE CRITICAL REPRODUCIBILITY EVENT (exp 49): a fresh reproducer with identical code, model hash, cell, surface, and continuation gave P(1)=0.610 vs the stored P(1)=0.005 - cause: a transformers library update (5.12.1) changed attention computation. The AUC 0.918 result and the digit-copying phenomenon from exp 49 are unreproducible artifacts. LESSON (entered into the evidence gates): pin library versions in every result file.
- THE DEEPER NEGATIVE (exp 58, 500 readouts): NOTHING transfers across questions - cross-question TV AUC 0.561 (chance), margin AUC ~0.49 (chance). No single-question measurement predicts behavior under a different question. This is a deeper negative than "omega fails."
- MECHANISM (post-corrections): UNRESOLVED. "Continuation-dependent sensitivity" is the accurate description. Remaining alternative explanations per review: readout redistribution rather than recovered information; ordinary access to retained history; position/tokenization/query interaction; softmax saturation; selection effects.
- Systemic corrections applied: the eligibility filter (omega defined only on pairs that start close - d_pre <= tau; several headline claims were eligibility artifacts and were revised); the boundary bug (#38 progressive-buildup narrative withdrawn - continuations without trailing newline got glued to the query); the safety paradox (withdrawn, n=6 p=0.54 bugged panel); the bifurcation predictor (downgraded: greedy-decoding/11-bin artifact pending token-persisted rerun).

PART F - CAMPAIGN II: CUMULATIVE CAPABILITY ACQUISITION (S1-S65, paused)

Question: can a system combine useful computations from successive pretrained models while retaining prior capabilities? 666 recorded forward calls over two days, nine probe blocks, zero building rounds (the measurement-to-artifact governance ratio fired honestly).

- S6: legacy plan exports ineligible (nonfinite/overwhelmingly zero).
- S21: four isolated conditional rectangles below registered tolerances at 8 forwards; audit found an additive preference class also fits - no conditional mechanism established.
- S22: the locked equal-rank two-donor combination LOSES to the retrospectively better individual donor on all four assays (8,610 variants) - a scoped negative with exact tie handling corrected by audit.
- S23 (proof-only): pairwise compatibility need not imply joint compatibility; compatible marginals need not identify new joint answers.
- S26 (audited e1044-e1045): the autonomous-continuation probe - primary pair recovery 0/16, second-site 2/16 vs cyclic control 3/16; the primary first proposal correct only 1/16, bounding attainable recovery below the 0.10 gain threshold. Closed on this support; weak proposals prevent a strong conclusion against competent composition.
- S58 (the modest positive): correct specialist pseudo-labels improve a frozen-feature species readout after teacher removal (2400 head-optimization steps); soft targets offer NO advantage over hard labels. The teacher's restricted labels match all 120 training truths. Scoped: a limited species readout, not new backbone knowledge.
- S65 (final diagnostic): supplied-fact reading works (3/3) but prerequisite composition is 0/3 - a scoped interface limitation, not missing knowledge or intrinsic composition failure. The natural-question prerequisite remedy was designed and never run (the three direct capital questions for Syria, Malawi, South Korea - a bundled interface remedy, not a causal wording isolation).

PART G - CAMPAIGN III: SPECTRAL / FRAME ANALYSIS (September 17-18, closed)

Frame-theoretic analysis of 3 trained models (SmolLM2-135M d=576, Qwen3-0.6B d=1024, Phi-1.5 d=2048):

1. Real models are 100x-10,000x from the tight-frame ideal: embedding condition numbers 640 (Phi) to 11,971 (SmolLM2); MLP conditions 180-25,888. The formal superposition theory's assumptions (nearly tight, low coherence) are massively violated.

2. Coherence 15-30x the Welch bound (0.75-1.00 vs 0.02-0.04).

3. Eigenspectra follow power laws (exponents -0.35 to -0.90), not Marchenko-Pastur or uniform.

4. Embedding condition DECREASES with d (spectral democratization with scale).

5. Gate/up asymmetry universal (gate_proj higher condition, lower effective rank - sharper selection vs broader computation).

6. AMPLIFICATION GRADIENT (entry 118 overturning entry 92): controlled fixed-magnitude perturbations into individual eigenspaces show TOP eigenspaces are 3-14x MORE sensitive (beta +0.25 to +0.41). Entry 92's "tail vulnerability" was a multi-layer processing artifact - the model absorbs top-eigenspace perturbations through compensatory dynamics, leaving tail residuals.

7. Information-theoretic eigenspace priority (entry 94, later scoped by review to selected tokens only): selected rare morphological tokens had greater top-eigenspace energy than selected common tokens.

8. Power laws require directional structure (entry 95): Monte Carlo - random directions with information-weighted norms produce FLAT spectra; the power law requires training organizing tokens into correlated directions.

9. First-principles (entry 101): a single-layer embedding trained with SGD on Zipfian tokens produces power-law eigenspectra; uniform tokens -> flat. Exponent controlled by (Zipf s, n/d ratio) - the FALSIFIED prediction was monotone alpha(d); alpha depends on n/d not d alone.

10. Spectral depth gradient (entry 112): MLP up_proj exponent steepens monotonically with depth (SmolLM2 rho=-0.891, Qwen3 rho=-0.649) - a generalization-to-specialization gradient; attention QK/OV show no depth trend.

11. Complementary specialization (entry 113, scoped): hidden-state spectra flatten with depth (rho=+0.871) while weight spectra steepen (-0.891) - oppositely moving slopes in one model and sampled corpus; causal/functional significance unestablished.

12. Descriptive-not-predictive (entry 115): spectral exponent does NOT predict layer sensitivity to ablation (U-shaped: first/last layers critical regardless). Sensitivity is position-driven.

13. D_k insufficiency (entry 103): permuting up_proj rows preserves ALL Gram invariants but destroys MLP output (139% error); function-preserving control has zero error - separation 5.5 million to 1. Any invariant factoring only through W^T W is blind to the row pairing SwiGLU needs.

14. The pairing paradox: Gram operators show population alignment (D1_norm ~ 0.22 constant across a 3.5x scale ladder); individual row pairs show no alignment (cos ~ 0); yet the specific pairing determines function. Information at no single natural level of analysis. Corrected framing (entry 107): hidden-unit gauge symmetry, not a tensor phenomenon.

15. Spectral-composition bridge REJECTED (entry 110): spectral structure and compositional pairing are INDEPENDENT - training does not jointly organize them. Permutation damage predicted by ||D*delta*U|| (the down-up product), not gate-up coupling.

16. Tensor-composition line CLOSED (entry 109): weight-only tensor predictor R2 0.94 vs DU predictor 0.95 - gate structure adds NOTHING for damage prediction.

17. MP-deviation analysis KILLED: eigenvalue spikes above Tracy-Widom exist (1-3 per model) but add NO predictive power over standard PCA; the 69x24 behavioral transfer data is intrinsically rank-3 - three architecture-family factors capture 79-97% of variance. The dataset is exhausted for high-d analysis.

18. Three theory-first cycles KILLED (post-review): concentration-of-measure bound on behavioral rank (base/instruct identical attention routing despite 2x behavioral PR); weight spectral bridge (identical weight spectra, 98% different behavioral PR - no correlation, all p>0.25); Jacobian derivation (value/routing decomposition backwards, the one sharp prediction falsified 6/7 models).

Campaign verdict: CLOSED. The d_add_zero dissociation survived (a perturbation appending +0 that touches zero original positions does NOT behave like an inert suffix - computational relevance, not position, determines sensitivity), and the R31 disambiguation (10x effect from surface-dependent parse). Power-law spectral origin left open - carried into Campaign IV.

PART H - CAMPAIGN IV: THE HD-MATH THEOREM PROGRAM (September 18-23, paused mid-confrontation)

The vocabulary: features f_1..f_k in R^d; Sigma covariance; PR(Sigma) = (tr Sigma)^2/tr(Sigma^2) the participation ratio; A_2 = sum_{i<j}(f_i.f_j)^2 the aggregate interference; the QSAE loss L(W) = sum_i p_i ||W^T W e_i - e_i||^2; softmax compression of a conditional kernel P(j|i) through rank-d logits; delta = z - log q the logit residual; C_i the fourth-moment contraction operator (for Gaussian f_i: C_i[x] = tr(x Sigma_i)Sigma_i + 2 Sigma_i x Sigma_i by Wick).

H.1 The spectral-law arc (HP-A/HP-B)

Design-gated twice before any run (the first adversarial review returned NO-GO with five required fixes: mean-field stationarity corrections, P2 n-vs-d mismatch, P3 wrong ODE form, operational falsifiability gaps, calibration leakage).

- THEOREM (sparse-exact optimum): the QSAE global minimum is orthonormal storage of the top-d usage tokens with the tail dropped; L_min = tail usage mass; the optimal frame operator is I_d (completely flat spectrum, condition 1). Proof: L = tr((G-I)P(G-I)) with G = W^T W PSD rank<=d; on ker G the factor acts as -I, so L >= tr(P|_kerG), minimized over kernel choice. Verified: L-BFGS from 5 inits at six (s,n,d) configurations - loss minus tail-mass bound = 0.000000 at all six; nonzero norms = d exactly.
- Consequence: power-law spectra cannot come from the quadratic objective's optimum - it anti-mixes.
- THE RANK-2 COLLAPSE: tied-softmax training on unigram Zipf data (n=1024, d=64, 192k steps, 3 seeds) collapses to rank 2 - top-2 eigenvalue share >= 0.984 at EVERY checkpoint from t=3k; norms uniformize (spread 1.9->1.4); curved fits beat power law 12/12; measured alpha >> the norms-only control (0.22). Construction: w_i = u + alpha_i e with alpha_i ~ log p_i realizes the target in rank 2. "Zipf usage + training produces power-law spectra" is an estimator artifact.
- ESTIMATOR VALIDITY (the measurement contribution): the standard log-log OLS spectral exponent carries (gamma, profile)-dependent bias up to +/-0.2 on TRUE power-law inputs (gamma=4 inflates shallow; gamma=16 deflates steep); FLAT inputs read as alpha ~ 0.40-0.45 at gamma=4 (threshold 0.35 fails); bulge inputs manufacture slopes up to ~12 but a quadratic-in-log alternative correctly rejects them (delta R2 0.13-0.28). Consequence: spectral exponents are comparable only within fixed (gamma, window) - prior cross-scale comparisons (including entry 101's) are partly instrument artifacts.
- The optimizer grid: 12/12 completed runs converged with flat endpoints (the theorem's prediction on every run); the heavy confirmatory arms were foregone with review-backed provenance (~7h/run economics; reviews 3-4 had demoted them as theorem-fixed).

H.2 The aggregate-interference arc (HP-E)

- THEOREM (isotropic): E[A_2] = k(k-1)/d exactly; Var(A_2) = 2k(k-1)(2d-2)/(d^2(d+2)) exactly (all shared-index pair-square covariances vanish identically by conditional independence + exact unit norms); two-phase frontier k* = Theta(sqrt d) with Chebyshev bounds. Pre-registered crossing confirmed (A_2 0.87-0.97 at k=sqrt d vs 14.8-15.7 at 4 sqrt d).
- THEOREM (participation ratio): for unit-energy features with covariance Sigma, per-pair interference = tr(Sigma^2)/(tr Sigma)^2 = 1/PR(Sigma) exactly; the frontier is k* = Theta(sqrt(PR(Sigma))); for power-law spectra with 1/2 < alpha < 1: k* = Theta(d^(1-alpha)) - a steeper trained spectrum taxes computation by d^(alpha-1/2) vs isotropy. Scope (adversarially enforced): expectation-level for the specified quadratic model; NOT a decoder/task capacity theorem.

 COVARIATE INSUFFICIENCY (pre-registered, decisive): Gaussian-direction and coordinate-sparse ensembles with IDENTICAL Sigma share the mean (few %) but differ 21x in cv (1.64 vs 0.076 at d=512, k=64, alpha=0.5) and 14x in spread. The spectrum sets the frontier's mean; the direction distribution's localization sets its sharpness - invisible to every second-moment tool.

 TWO-TERM LOCALIZATION LAW: for coordinate-sparse directions on diagonal Sigma = diag(sigma_r): cv^2 ~ 2(Psi_1 - 1)/k^2 + 2(Psi_2 - 1)/k, Psi_1 = d^2 tr(Sigma^4)/(tr Sigma^2)^2, Psi_2 = d tr(Sigma^4)/(tr Sigma^2)^2 - exact at alpha=0 in BOTH terms (the sqrt(2d)/k law reproduced to 2%; the shared-index term vanishes exactly where predicted); no free constants. Gaussian directions collapse to 2/k.

 THE EXACT BIRTHDAY COVARIANCE: for S = sum_r w_r C(N_r,2), N multinomial(k, uniform): E[S] = (1/2)(k)_2 p^2 W_1; Var(S) = (1/4)[W_2 V + (W_1^2 - W_2) C] with V = (k)_4 p^4 + 4(k)_3 p^3 + 2(k)_2 p^2 - (k)_2^2 p^4 and C = -(k)_2(4k-6)p^4 - the NEGATIVE fixed-k correlation the Poissonized approximation misses. THE REFUTATION THAT PRECEDED IT: the Poissonized (independent-channel) prediction was pre-registered and FAILED 2/12 (measured cv^2 dropped ~10x per 4x lambda while the formula is lambda-free). The exact formula validated 12/12 at ratios 0.76-1.09, zero free parameters. Law class governed by lambda^2 PR_W; asymptotic normality from the occupancy CLT (Holst 1972; Karlin 1967) when the Lindeberg/max-cell condition holds; compound-Poisson near the birthday threshold.

 HIGH-PROBABILITY FRONTIER: Chebyshev on the exact variance - the two-phase crossing holds w.h.p. beyond the (Psi_1, Psi_2) burn-in; Gaussian directions need only k >= O(1).

H.3 The compression arc (HP-D, the Orlicz program)

 REPRESENTABILITY: the planted-block kernel has exactly B distinct conditionals -> centered log-kernel rank exactly B (verified to 1e-15); exact zero-excess representation iff d >= B. Untied logits Z = W^T H range over all rank-d matrices; softmax rows are shift-invariant.

 FISHER-METRIC COMPRESSION LAW: excess CE ~ (1/2) x tail Fisher energy beyond rank d (Eckart-Young in the Fisher metric - entries weighted by sqrt(W_b P_bj), context mass x conditional mass). Validated: tight (1.08-1.28x) far below representability at s=0.5 (4-8 points within 1.2x); the inf-norm bound overestimates 10-30x because KL hides residual where P_ij is small - the Fisher metric is the tight second-order object.

 THE s=1 GAP: at steep usage the optimum settles (12 seeds within 0.0005) at 6.2x/4.7x the Fisher prediction - GENUINE large deviation (not optimizer error; the alternative explanation would need 12 seeds all within 0.0005 of a wrong value).

 THE EXACT CGF IDENTITY: KL(q||softmax(z)) = log E_q[e^delta] - E_q[delta] - the CUMULANT GENERATING FUNCTION of the logit residual. Proof: KL = -H - <q,z> + log sum e^z; substitute z = log q + delta. The Fisher law is exactly the kappa_2 term.

 THE ARGIN CHANGE: at the settled s=1 optimum, the quadratic component alone (0.0966) is 7x the Fisher-optimal value (0.0139) and exceeds the true total (0.0859); kappa_3 = -0.005 partially cancels - the optimum DELIBERATELY accepts 7x the variance to buy negative-skew cumulant structure. Softmax compression is a different mathematical object from low-rank approximation: the argmin changes qualitatively, not perturbatively.

 ENVELOPE DEGENERACY: the moment problem "minimize KL at fixed residual variance" is degenerate - rare huge negative jumps (probability eps->0, magnitude ~sqrt(v/eps)) carry the variance while E_q[e^delta] -> 1: inf KL = 0 at ANY fixed variance. Polynomial moments of delta do not govern KL compression; the one-sided exponential moment does (Orlicz scale). This explains all three measured regimes at once: Fisher agreement where deviations are small; the 6x gap where the optimizer reshapes the exponential tail; super-efficiency near representability where positive-tail mass relocates to vanishing-q entries.

 FIRST-ORDER STRUCTURE (verified): the optimal rank-d R* satisfies Eckart-Young-type orthogonality - the weighted gradient field W_b(softmax(r*_b) - q_b) vanishes ON the row space (lives in the complement block) - verified at off-space gradient energy = 1.0000 at the settled optimum. (The statement's containment direction was initially reversed; the numerical check exposed it; corrected in place.)

 STRICT LOCAL MINIMALITY: evidence-grade - Lanczos Hessian on the GL(d)-quotient gives lambda_min = 1.18e-5 > 0 (four orders below lambda_max 0.252 - a genuine near-flat valley explaining the 0.0005 seed reproducibility) but at the finite-difference noise boundary. The naive full-space certificate was challenged for gauge modes and downgraded.

 DUALITY IMPOSSIBILITY CLUSTER: three Eckart-Young-style duals fail structurally - (a) SVD-based dual: the inner linear problem on the unconstrained rank-d variety is unbounded (-infinity); (b) Donsker-Varadhan dual inherits it; (c) the lambda-regularized dual is vacuous at any lambda preserving the problem (the optimum carries ||R||_F^2 ~ 1.1e4). ROOT CAUSE: KL geometry has no norm ball - logits scale freely because only entropy differences matter. Optimality theory must be simplex-native.

 THE HIERARCHICAL FAMILY: for the level-mixture kernel P(j|i) = (1-rho)p_j + rho sum_ell pi_ell 1[same level-ell ancestor]: the centered log-kernel has rank exactly 2^L - 1 (nested-bump orthogonality: same-level bumps disjoint, cross-level orthogonal after centering by the tree property); bundle-scale law sigma_ell ~ sqrt(m_ell) Delta_ell (geometric under geometric pi); the excess curve is a recursive double tail (level envelope x within-bundle Zipf block-mass tails). EMPIRICAL (after a construction bug was caught and fixed - the kernel was not row-stochastic): smooth log-convex decay to EXACT zero at the representability rank 32 (both rho arms);

Fisher envelope tight 1.08-1.28x far below; super-quadratic near the knee (to 0.02x). The staircase prediction REFUTED (smooth, not stepwise). THE STEEPER-PI PROBE (decisive, pre-registered): the family CANNOT reach the CGF regime at any steepness (max ratio 1.61 vs the 1.8 threshold at pi=1/8, rho=0.7) - the shared-root floor bounds likelihood contrasts. Scoped conclusion (review-corrected): within the tested constructions, nested kernels live in the Fisher regime - construction-forced, not a general overlap law.

- THE OVERLAP-PARAMETER ARC (HP-F): opened to replace qualitative "disjointness" with a quantitative gate; the first falsifier design was caught by review BEFORE running (the eps-floor bounds Lambda at fixed vocabulary; the p_min->0 route confounds all mixing levels); the residual-level CGF remainder bound was derived and verified (cube-exponential Taylor form; R = KL - Var/2 = -0.0107 at the settled optimum, bound 0.101 holds with 9.5x slack - later scoped as a generic loose Taylor remainder, not regime identification); the matched-kernel construction (equal Lambda + Fisher spectrum, different CGF tails - the identifiability test of kernel-level vs residual-level gating) was honestly deferred (the naive column permutation gives a symmetry null, not a discriminator). Superseded by HP-G per review-10.

H.4 The general theorem (HP-G, the unification)

- G1 (mean, triangular-array general): E[A_2] = sum_{i != j} tr(Sigma_i Sigma_j) - second moments only.
- G2 (variance via the contraction operator): the shared-index covariance is 2 tr(Sigma_j Sigma_i Sigma_l Sigma_i) (Gaussian instance; general form: <Sigma_j, C_i[Sigma_l]> minus product of means); the pair-variance term E[(f_i.f_j)^4] - E^2 = 2 tr(Sigma^2)^2 + 6 tr(Sigma^4) (homogeneous; verified at ratio 1.0032, 400k reps - a wrong first candidate caught by the Monte Carlo before recording).
- G3 (counting, derived on paper): Var(A_2) = C(k,2)[2 tr(Sigma^2)^2 + 6 tr(Sigma^4)] + 4k C(k-1,2) tr(Sigma^4) - CERTIFIED at three configurations by brute force (0.9994 / 1.0021 / 1.0031). The explicit L2 criterion: cv^2 = [2 + 6 t4/t2^2]/C(k,2) + 4kC(k-1,2) t4/[C(k,2)^2 t2^2]. An ordered/unordered algebra slip in the first criterion was CAUGHT BY THE VALIDATION DATA ITSELF (the stored gauss-arm data matched the unit-norm law, not the Gaussian law; the corrected criterion then matched: 0.0673 vs 0.0708). The intermediate "norm-fluctuation exhibit" was retracted (the genuine effect is a factor sqrt(2), not an order of magnitude).
- G4 (universality classes): the diagonal probe R = sum_a E[f_a^4]/sum_a (E f_a^2)^2 equals 3.0 for Gaussian (dimension-free - the universality center) vs ~d for coordinate-sparse (dimension-growing). THE UNIVERSALITY CROSSOVER LAW: for mixture directions sqrt(1-eta) Gaussian + sqrt(eta) sparse, the Gaussian class boundary obeys ETA*(d) ~ d^(-1/2) - departure when eta^2 d ~ 1. CONFIRMED BY PRE-REGISTERED BRACKETING at d = 512 (eta* in (0.30,0.35]), 2048 and 8192 (both (0.15,0.25]), and the pinning cell d = 32768 (within at eta=0.08: cv 0.0389; exceeds at eta=0.12: 0.0637 vs line 0.0630) - the d^(-1/2) bracket. This is the program's first dimension-asymptotic phase law: direction ensembles within eta of Gaussian (in the contraction-operator sense) share the sharp interference frontier; beyond it the frontier broadens. The formal Lipschitz proof is a written skeleton (both variance terms are linear in C_i; the mixture perturbation carries eta(1-eta)d diagonal-probe mass; the k-power bookkeeping must reproduce the measured law) - OPEN.
- G5 (validation against existing data): performed, and caught the algebra slip above - the validation data was the instrument that certified the corrected criterion.

H.5 The out-of-family confrontation (paused mid-run)

Corpus assets: WikiText-103, ~120M tokens per model, own tokenizer, top-8192 vocabulary (kept mass 0.889-0.906), +-2 window co-occurrences as CSR (nnz 9.2-9.8M per model) - all three models (SmolLM2, Qwen3, Phi-1.5) staged.

THE FROZEN PREDICTION (committed before measurement): the Fisher-tail excess-CE curve for the SmolLM2 kernel, usage-probability-weighted, alpha-smoothed 0.1, d grid 8-190: d=8: 1.795, d=16: 1.238, d=32: 0.820, d=64: 0.484, d=96: 0.308, d=128: 0.183, d=160: 0.083, d=190: 0.005.

The measuring run (rank-d CE optimization on a 2048-context usage-weighted subsample): completed d=16 (excess 6.01 vs full-kernel curve 1.24 - UNINTERPRETED: the honest comparison is against the subsample's own Fisher tail, computed at run end, not reached); d=64/128 pending; paused by the owner with the runner idempotent (~3-4h). Scoring rule frozen: direct curve comparison, no exponent refitting. The gauge caveat stands: factor spectra are non-identifiable from kernel spectra (AB^T = (AS)(BS^-T)^T); the loss curve and logit operator are the gauge-safe observables.

PART I - THE COMPLETE FAILURE/LESSON REGISTER

Belief that died	How it died	Transferable lesson
Nested-arithmetic capability claims (legacy)	Termination-under-cap artifact	L0: report termination beside every accuracy

"Zipf + GD produces power-law spectra (RESOLVED)"	Sparse-exact optimum + rank-2 collapse + estimator manufacture	L1: verify the instrument on known inputs before reading structure out; a fitted slope is not a mechanism
Poissonized birthday prediction	Failed its own pre-registered test 2/12	L2: conditioning structure can be leading-order physics, not a correction
Variance-KL discount law	Moment problem degenerate (rare negative jumps)	L3: polynomial moments do not control KL; check the variational problem before expanding
Three duality certificates	Unbounded inner problems; KL has no norm ball	L4: Eckart-Young duality needs norm geometry; simplex problems need simplex-native theory
"Capacity bridge" (4 wording instances)	Expectation-level onset != decoder/task capacity	L5: name decoder, task, error criterion before the word "capacity"
Norm-fluctuation order-of-magnitude exhibit	Validation data caught an ordered/unordered slip	L6: certify the FULL derived quantity against data, not just components
HP-F falsifier sweep	Review caught pre-run: eps-floor bounds Lambda; sweep cannot isolate its variable	L7: prove the manipulation moves X independently of confounds before testing "X gates Y"
"Disjointness gates CGF" (general)	Two constructions cannot ground a general law	L8: replace qualitative gates with quantitative parameters measured on the objects
Corpus spectral-inheritance matching	Factorization gauge freedom	L9: predict gauge-invariant observables (loss curves), or fix the gauge by objective
Strict-local "CERTIFIED"	GL(d) gauge modes unquotiented	L10: quotient the gauge group before curvature statements
Hierarchical staircase knee	Non-stochastic kernel construction bug + smooth true curve	L11: assert normalization invariants before ANY derivation-support numerics
Synthetic-family tunnel (diagnosed 3x)	Families' built-in spectra force their conclusions	L12: the contribution must be asymptotic laws tested outside the suggesting family
Quadratic-moment internalism (diagnosed)	Reviews 10-11: exact variance alone is classical U-statistic machinery	L13: the new content is the critical-regime classification and universality - dimension-asymptotic laws, not repackaged Wick/Chebyshev
Omega consequence AUC 0.918 + digit copying	Transformers library update changed attention; unreproducible	L14: pin library versions in every result file; a fresh reproducer is part of the claim
Omega cross-question transfer	At chance (AUC 0.561) - nothing transfers across questions	L15: single-question measurements do not predict behavior under other questions - a deeper negative than "omega fails"
Safety-comment paradox (omega)	n=6, p=0.54, bugged panel	L16: small-n winner's-curse panels cannot carry headline claims
Bifurcation predictor (omega)	Greedy-decoding/11-bin artifact	L17: readout and decoding choices are load-bearing; test them before dynamics claims
Phi-1.5 "no gradient"	Readout artifact: trailing space flips P(digit) 0.001->0.980	L18: model-specific admissible readouts govern cross-model comparison
Copy-vs-compute control	Cell-schema mismatch (labels factually false)	L19: verify the data's labels match the analysis schema before entering results in STATE
5.6x instruct gap (Qwen3)	Definition artifact (two files computed different metrics)	L20: like-for-like requires the same metric definition, not just the same model
"Position dominates content"	Entirely the A/B after-block cohesion already known	L21: content identity is undetectable in this design; position vs header-adjacency not separable
Instruction-tuning "selective reshaping"	SmolLM2 replication went the OPPOSITE direction	L22: n=1 architecture pairs cannot carry directional claims; replicate before interpreting
R20 row-permutation interaction test	Null operation (row-perm invariant); all p-values were roundoff	L23: a null_std at 1e-20 is a stop signal, not a result
Prefix-form mediator	Column main effect, not diagonal (R25)	L24: check the design's contrasts before narrating the diagonal

Entry 94 "optimal coding"	Selected tokens only; no optimality benchmark	L25: observational wording for observational data
Entry 100 "universal coupling budget"	3 confounded checkpoints	L26: scale ladders confound width/depth/training unless controlled
HP-C1 exponent matching (proposal)	Gauge non-identifiability (review)	L9 (restated): the loss curve is the observable

PART J - PROCESS KNOWLEDGE (how the program ran)

 THE REVIEW PROTOCOL: a goal-aligned adversarial reviewer on a fixed cadence - attacking method and strategy to ensure the best path to the goal, NEVER closing the mission. Every review caught at least one real defect (sign errors, reversed containment, non-stochastic kernels, overclaim patterns, in-family verification traps, pre-run design flaws). The program's rule: apply corrections verbatim in the same session; record with provenance; an audit without "should this continue and is it the highest-leverage thing?" is not an audit.

 THE PRE-REGISTRATION PATTERN: freeze predictions in a committed file before the run; validate with zero free parameters; score honestly - two registered predictions were refuted by their own tests and both refutations led directly to the correct formulas.

 THE TUNNEL DIARY (the program's own overconfidence pattern, named by reviewers across three waves): spectral-exponent tunnel (fitting slopes), synthetic-family tunnel (constructions that force conclusions), quadratic-moment internalism (repackaging classical tools). The countermeasure each time: dimension-asymptotic laws tested outside the suggesting family.

 THE CHEAPEST-CONFOUND RULE: before accepting any class-level claim, ask what the simplest construction-forced explanation would be (task degeneracy, readout artifact, gauge freedom, normalization bug) and test it first.

 EVIDENCE GATES: termination beside accuracy; cheapest-remedy arm first; fixed-input null for sampling claims; held-out tasks for headline numbers; library versions pinned; propagation of every downgrade to every public surface in the same session.

PART K - STATE AT PAUSE AND OPEN OBLIGATIONS

Priority order: (1) the frozen confrontation (resume: python experiments/run_hpf_confrontation.py, idempotent; score against frozen_prediction_smollm2.json; no refitting); (2) G4-formal - the Lipschitz bookkeeping proving eta* ~ d^(-1/2); (3) universality-class separation (identical 2nd+4th moments, different limits - or prove impossible); (4) the same frozen-test protocol on the Qwen3 and Phi-1.5 kernels; (5) the bridge back - intervene on a real model's representation spectrum and measure superposition capacity against the d^(1-alpha) frontier.

What is honestly NOT established: no theorem tested by intervention in a live transformer; every theorem scoped to its specification (quadratic readouts, softmax factorizations, planted/hierarchical kernels, named ensembles); several claims expectation-level or evidence-grade; all review internal (one external CLI reviewer), not independent peer review.

> *Standing judgment: the direction is alive and productive. Five theorem families stand with validated or certified cores; the universality crossover is a genuine dimension-asymptotic phase law of the kind the mission demanded - it does not exist at low dimension at all. The distance to the terminal conditions remains large. The next decisive event is the confrontation between the compression theory and a real corpus it did not design. The discipline - pre-registration, zero-parameter validation, goal-aligned adversarial review, honest retraction - is itself a transferable artifact: it caught every error in Part I before anything false entered Parts H-G.*